Acta Astronautica 220 (2024) 185–196

Contents lists available at ScienceDirect

Acta Astronautica

journal homepage: www.elsevier.com/locate/actaastro

Importance and challenges of integrating BLSS into ECLSS

Megan Kane *, K.C. Shasteen

Department of Biosystems Engineering, University of Arizona, Tucson, AZ, 85721, USA

ARTICLE INFO

Keywords:
Bioregenerative life support
Environmental control and life support
Advanced life support
Crewed space exploration
Controlled environment agriculture

ABSTRACT

Bioregenerative Life Support Systems (BLSS) are a key facet of plans for long term habitats on other celestial bodies. With the Artemis program headed to the moon and SpaceX pushing towards Mars the roadmap is about 10–20 years. Therefore it is vital to consider how BLSS will work with the traditional physicochemical Environmental Control and Life Support Systems (ECLSS). Biological systems are complicated. Their inputs and outputs cannot be turned on and off the way physicochemical systems can. Therefore the accurate monitoring and prediction of these systems is fundamental to integrating the BLSS into the life support system as a whole. This paper will summarize the history of BLSS research, the system requirements for integrating a BLSS into the ECLSS, and the research needed to meet these requirements. As the world looks towards the future of humans living on other celestial bodies, there is a lot of work to do to support keeping them safe and healthy. Bioregenerative Life Support Systems have potential to provide massive support, if they are effectively integrated with the physicochemical systems.

1. Introduction

In order to establish and maintain a permanent human presence off-world, it is necessary to fulfill all the physiological needs for people (and often other lifeforms) to perform tasks, rest, and maintain their health and wellbeing. This is a very broad definition of life support. Generally, a life support system will have potential interfaces with air, waste, water, crew, environmental monitoring and control (EMC), extravehicular activities (EVA), food, habitation, in-situ resource utilization (ISRU), medical, power, propulsion, radiation protection, and thermal systems [1]. Often a subset of these interfaces are managed by an Environmental Control and Life Support System (ECLSS), which is an automated system of regenerative life support hardware that provides useable air, water, and habitable volume through artificial means. A Bioregenerative Life Support System (BLSS) is a type of ECLSS which regenerates system capacity via biological rather than strictly chemical, mechanical, or physicochemical (PYC) processes.

This paper will explore how bioregenerative life support (BLS) fits into the long-term growth of space outposts and colonization and will offer informed suggestions about what direction future research must take to best achieve these objectives.

First, a brief history of BLSS development will be introduced, the system requirements of a BLSS will be defined, and, finally, the benefits and challenges of various future research options will be explored.

2. Development of BLSS through today

A great deal of foundational research has been done on BLS starting with closed system *Chlorella* algal cultivation in the 1950s and 60s. A host of other projects followed such as Bios-3, Biosphere 2, Biomass Production Chamber (BPC), South Pole Growth Chamber (SPGC), Mars Lunar Greenhouse (MLGH), Yuègōng-1 [Lunar Palace-1] (天宫1), Micro-Ecological Life Support System Alternative (MELiSSA), Closed Ecological Experiment Facility (CEEF), Eden ISS, and many experiments from the University of Guelph using hypobaric plant test chambers. An excellent review of many of these systems is covered in Wheeler, 2017 [2].

All of these projects grew higher plants and or algae as a means of gas exchange: i.e. to sequester carbon dioxide (CO_2) and produce oxygen (O_2) within the growing space. Some of the projects, such as Bios-3, Biosphere 2, CEEF, and Yuègōng-1, involved enclosing humans (up to two years) and, additionally, produced calories for the crew by growing plants. Yuègōng-1, CEEF, and MELiSSA enclosed animals (mealworm, goat, and rat, respectively), Biosphere 2 had many species in various ecologies, and Guelph experimented with pollinating insects (bumble bees) [3]. Some projects achieved greater rates of atmospheric closure (the degree to which resources were recycled without external inputs). Both Bios-3 and Yuègōng-1 reported very low atmospheric leakage rates of 0.020–0.026 % per day by volume [4] and 0.04 % per day, respectively [5], while BPC was operated in various configurations and had

* Corresponding author.
E-mail addresses: megankane@arizona.edu, megan@megankane.com (M. Kane).

https://doi.org/10.1016/j.actaastro.2024.04.016
Received 15 January 2024; Received in revised form 24 March 2024; Accepted 9 April 2024
Available online 10 April 2024
0094-5765/© 2024 Published by Elsevier Ltd on behalf of IAA.

Acronyms/abbreviations

BLS	Bioregenerative Life Support
BLSS	Bioregenerative Life Support System
BPC	Biomass Production Chamber
BVAD	Baseline Values and Assumptions Document
DLR	Deutsches Zentrum für Luft und Raumfahrt
CEA	Controlled Environment Agriculture
CEEF	Closed Ecological Experiment Facility
CM	Crew Member
COTS	Commercial Off-The-Shelf
CT	Crew Time
DWC	Deep Water Culture
EC	Electrical Conductivity
ECLSS	Environmental Control and Life Support System
ELS	Exploration Life Support
EMC	Environmental Monitoring and Control
ESM	Equivalent System Mass
EVA	Extravehicular Activities
HI	Harvest Index
HVAC	Heating, Ventilation, and Air Conditioning
ISRU	In-Situ Resource Utilization
ISS	International Space Station
LED	Light Emitting Diode
MLGH	Mars Lunar Greenhouse
MELiSSA	Micro-Ecological Life Support System Alternative
NCT	Non Crew Time
NFT	Nutrient Film Technique
pH	potential of Hydrogen
PYC	Physicochemical
ROI	Return On Investment
SAM	Space Analog for the Moon and Mars
SIMOC	Scalable, Interactive Model of an Off-World Community
SMAC	Spacecraft Maximum Allowable Concentration
SPGC	South Pole Growth Chamber
TPCA	Top Projected Canopy Area
VCD	Vapor Compression Distillation
VOC	Volatile Organic Compound

leakage rates of about 5–10 % per day when closed [6]. Notably, the University of Guelph experimented with growing plants and the kinematics of pollinator flight at reduced pressures and in alternate atmospheric compositions (e.g. increased/decreased partial pressure of O_2), with interesting results showing growth in a pressure as little as 10 % of sea level and flight at reduced pressure provided partial pressure of O_2 was sufficient [3,7]. Biosphere 2 attempted very large scale ecosystem simulations, and the MELiSSA project continues to experiment with integrating growth systems for higher plants, microorganisms, and crew in a smaller scale, more portable artificial ecology. Almost all the projects primarily used hydroponics as a way of eliminating the complexities of and gas exchange related to soil microbes and soil nutrient cycles. Some of the many important factors these projects attempted to characterize were power usage rates, plant productivity, gas exchange, volatile organic compound (VOC) evolution, trace nutrient/contaminant concentration, degree of closure, water usage rate, leakage rate, and microbial cultural evolution.

Among the most important findings of this foundational research were that, during long enclosures, nutrients such as sodium (Na) and potential contaminants such as lead, chromium, nickel, aluminum, titanium, and zinc (Pb, Cr, Ni, Al, Ti, Zn) could build up within the nutrient solution. Some of these elements were from crew consumables entering the water cycle (like soap and toothpaste) while others (like lead) were from equipment, i.e. solder used in the construction of the plumbing system [4]. Likewise, the enclosed atmosphere could become contaminated by silanes, flame retardants, and plasticizers which would off-gas from sealants, electronics, structural polymers, and coverings [8]. These could potentially cause issues by fouling air filtration systems and affecting the replacement rates on filtration consumables (such as activated charcoal, resin, and catalyst beds). From such findings it has become clear that careful material selection and long off-gassing periods are required for flight hardware.

Another source of possibly hazardous or obnoxious VOCs that have been characterized were biogenic contaminants such as ethene (a.k.a. ethylene) and isoprene to name a few [8–10]. These gasses can act as regulators of plant biological processes and, if their concentrations are allowed to grow unchecked, could affect the functioning of a BLSS. Similarly, plant growth, cooking, and decay related byproducts such as sulfides can cause odors and breakdown to potentially produce toxic effects for crew or catalytic filters [11,12]. Similarly, humans can produce hazardous gasses that can also build up in fully closed environments such as carbon monoxide (CO) [13 pp. 131–134]. Understanding the interplay of phylogenic, microbiogenic, anthropogenic, and technogenic gasses with crew, plants, microorganisms, and air filtration systems together with filter sparing and maintenance is critical to the sustainability of long term human enclosures. It is clear from previous work that volatile evolution is highly species dependent [10], so high fidelity (low leakage) enclosures and gas analysis must be done to integrate new types of plants into any kind of BLSS.

Many of these projects are ongoing in some kind of deployed form, such as SPGC, and some continue to do research, such as MLGH and MELiSSA. Other, newer efforts have also begun. For instance, Deutsches Zentrum für Luft und Raumfahrt (DLR) is designing growth systems for off-world, planetary bases using lessons from EDEN ISS [13]. At Biosphere 2, an analog environment called Space Analog for the Moon and Mars (SAM) is doing small scale pressurized enclosures with crew [14]. There, in addition to providing a research environment for others, they hope to validate a resource exchange model called Scalable, Interactive Model of an Off-World Community (SIMOC) [15]. This is an example of an important type of model that will be needed to plan and control the operation of a BLSS in conjunction with a PYC ECLSS, and these will be discussed in detail later.

Some interesting research is also on-going using cyanobacteria for both gas-exchange and soil remediation [16,17]. This approach could provide a form of ISRU, either as a means of producing useable soil locally or by extracting nutrients from the available regolith for hydroponic solutions.

Nutrient recycling is an important part of achieving high levels of closure. At 100 % closure, all resources are recycled within the system and no outside inputs are needed. Functionally, 100 % closure is impossible; there are always leaks, inefficiencies, or mission requirements that prevent it; however, quite high levels (>80 %) appear to be achievable for many resources. The carbon cycle, for example, is particularly important in contexts where crew are the only ready source of CO_2 for photosynthesis: such as the moon or deep space. In Bios-3, inedible biomass was incinerated in order to release sequestered carbon back into the cabin air, and urine was cycled back into the hydroponic solution [4]. In Yuègōng-1, a bioconverter used microbial action to decompose straw, human waste, and mealworm frass, but the solid waste product was not recycled. In Biosphere 2, gas exchange was originally complicated by off-gassing from structural materials, but, ultimately, both plant and human waste products would be digested by aerobic and anaerobic microorganisms via several waste treatment lagoons and then the product would be integrated throughout the larger ecology [18]. MELiSSA compartments I, II, and III use a sequence of microorganisms to break down solid wastes, first using aerobic

thermophiles, then photoheterotrophs, and then nitrifying bacteria [19]. Other research is currently examining the possibility of using edible mushrooms as a way of both releasing carbon and producing proteins from inedible biomass [20–22], though large-scale, fully-closed facilities for this research have yet to be constructed.

Many of the most recent innovations in indoor cultivation have come from the vertical farming industry. The growth of this industry and the greenhouse industry has helped fund development of light emitting diode (LED) lighting technology specifically for plants [23]. These LED lights are much more energy efficient and focus more of their spectrum into the photosynthetically active range than lights available during the original BLSS research. Accompanying research has since been developed around lighting schedules and light "recipes" which can further improve growth efficiencies. For example, in floriculture, recent research has found that, in some species, "monochromatic day-length extension" can be used to extend growing periods past "short day" limits without evoking a photoperiodic response [24]. Consequently, growing periods can be spread out over longer periods and less intense lights may be used. It is still not clear if such techniques can be applied to space crop species.

The role of airflow in plant growth is being studied for vertical farms, and CFD analysis is being used to explore facility level air system design [25–27]. New heating, ventilation, and air conditioning (HVAC) standards specifically for plant growth environments were recently published [28].

Automation technologies are currently being developed and deployed by vertical farming companies such as Plenty and OnePointOne. These include automated harvesting, overhead conveyors, and computer vision technologies for tracking plant status. Human in-the-loop AI and modeling work has also been developed for greenhouses which has been shown to outperform human growers for crops such as tomato [29].

3. Integrated system requirements ECLSS to BLSS

Research into Bioregenerative Life Support Systems has been primarily functional. With over 50 years of research and a full fledged spinoff industry in the form of Vertical Farms, it is now possible to examine those functions from a systems level. Considering the details of how to integrate a BLSS into a traditional ECLSS system is the next step.

3.1. Physicochemical ECLSS

On the PYC side, subsystems typically present in an ECLSS are water treatment and storage, food storage and preparation, waste management, air filtration, CO_2 scrubbing, pressure control, oxygen and inert gas storage (and/or production), air monitoring (and fire suppression), air circulation, and thermal management.

On the international space station (ISS), human waste urine is processed by vapor compression distillation (VCD), and the resulting distillates along with condensates (moisture mostly from sweat and breath) from the air system are then filtered by particle filters, resin beds, and a catalytic reactor [30]. A concentrated brine solution is output as a waste product. A water processing assembly can also optionally receive the water output from ISRU sources, such as a sabatier reactor (for reducing CO_2) or fuel cell (for generating power).

Food storage is nominally provided in the form of long-shelf-life, dehydrated food. In practice, however, there are several reasons to deviate from this approach. One, for longer missions, a dehydrated menu may lack enough flavor and variety and can create food boredom among the crew which may make it more difficult to hit calorie intake targets [31]. To counteract this, menus including frozen food stuffs have also been baselined [1]. Two, the practice of dehydration, and/or freeze drying, and special packaging requirements of space-foods adds substantial costs. As a way of reducing these costs, commercial off-the-shelf (COTS) food products have been increasingly used aboard ISS. One important problem with this approach is that COTS foods tend to be higher in sodium, and diets that include them can make it more difficult to hit the restricted sodium intake recommended to reduce bone loss [13 pp. 424–426]. For these reasons, the food storage subsystem can be as simple as shelves with pre-packaged meals ready to eat or as complicated as a freezer and refrigerator system. Food preparation systems, likewise, will [32]vary on how the food is stored and the amount of crew time available for cooking. Microwave ovens, ranges, food processors, crock pots, stove pots, pans, plates, trays, and silverware are all possible elements. In habitats using reduced pressure atmosphere mixtures [33–35], pressure cookers should also be expected.

Solid waste management is typically simple storage in sealed containers, though, in more complicated systems, there may be a drying or lyophilization step to reclaim moisture.

Air filtration systems typically involve elements such as particle screens, followed by activated carbon beds, and catalyst (typically platinum) coated charcoal beds [36]. These filters are usually driven by primary air circulation fans, but may also have their own dedicated fans.

Scrubbing of CO_2 has evolved over time. Original scrubbing technologies resembled scuba rebreathers and used moist hydroxide beds to chemically react with CO_2 in the air, depleting the filter over time [37, 38]. Currently on ISS, two CO_2 removal technologies are in use. One involves using zeolite filtration beds and thermal vacuum swing adsorption [39]. The second uses amine beds and thermal cycling to absorb and release CO_2 [40,41]. Both of these systems have the added benefit of instantly regenerating their filtration beds. Also, both are capable of concentrating CO_2 either for overboarding or for potential use as an input in oxygen production systems.

Pressure control is handled largely by the storage of oxygen and inert gasses and by providing an output path in case of overpressure. Any leakage that may happen over time, through the habitat structure or through EVA related airlock activities, will inevitably require make-up gas to restore pressure. An ECLSS should monitor the pressure and regulate release of this gas from storage. In the event of a decompression (or fire necessitating complete exhaust), it will require enough gas to completely replace the atmosphere. This system is also useful for controlling gas composition and preventing high O_2 situations which may present a fire hazard or low O_2 that might incapacitate the crew. Some pressure control systems may incorporate a "lung", which is either an adjustable volume or a pressurizable container that will temporarily hold/release excess gas during periods when, say, temperatures are rising/falling and driving changes in pressure.

Oxygen production is another role that may be provided by an ECLSS. Instead of releasing O_2 from storage, an electrolysis system or a chemical oxygen generator (typically a perchlorate candle) may be used to release O_2 into the habitable volume.

Air monitoring is an important function that acts as the eyes (or rather the nose) of the ECLSS. This system must monitor and maintain the composition, temperature, humidity, and alert crew to the presence of contaminants, fire, loss of pressure, or overpressure. At a minimum, the system should monitor O_2, CO_2, ammonia, CO, VOCs, coolants (whatever they may be), overall pressure, temperature, and humidity.

Air circulation is a critical function for maintaining proper atmospheric composition and reducing heat loads, especially in weightlessness where a lack of buoyancy will prevent heat convection. In gravitational contexts, when heat builds, buoyancy will naturally create a circulation which will spread the heat and circulate waste products in the atmosphere. In weightlessness, instead, these products will stagnate near the source, potentially creating hidden hazards. Even with circulation, CO_2 concentrations behind equipment have been observed to become potentially hazardous on ISS and Mir [13 pp. 131–134]. Circulation and thermal management is often tied into internal and external fluid loops, sensible and latent heat generated by the crew and equipment in the habitat air must be cycled past cold plates or through heat exchangers to remove heat and humidity and maintain a stable environment. Alternatively, sensible cooling can be supplied by vaporizing

cryogenic fluids such as O_2, as is the case for the SpaceX Dragon capsule. Moreover, a recent development in humidity control was made for the Dragon capsule which uses Nafion membranes to selectively remove water vapor from the crew space without needing to cool to the dew point [42].

3.2. BLSS elements

There are many types of BLSS as briefly mentioned above in section 1.1. This paper focuses on higher plant based BLSS and its functionality in gravitational contexts. As such, systems like Veggie and Advanced Plant Habitat, which were designed for microgravity, are not considered. Although they are similar in many ways, lack of gravity prevents the usage of NFT, which is the best explored nutrient delivery system for BLSS.

In practice, a BLSS should always be used in conjunction with a PYC ECLSS both for redundancy and for use during system startup when bioregenerative capacity is low. When accompanying a PYC ECLSS, a BLSS will require several additional subsystems: structure, lighting, irrigation, fertigation, growth zone HVAC, growth zone EMC, CO_2 injection, nutrient reclamation, VOC removal, O_2 removal, nutrient and seed storage, propagation area for seedlings, and a cleaning and maintenance space for equipment. Additionally, higher tech BLSSs will include subsystems for computer vision, moving shelving systems, and robotics (together these can be classified as mechanization or automation). Growing systems will also place significantly more demand on habitat power and thermal management systems, mostly due to the lighting system.

The structure of the growing space must provide enough growing area to meet crew dietary production goals for the mission, which depend greatly on how much food is expected to be resupplied from Earth. Based on the NASA Baseline Values and Assumptions Document (BVAD) from 2022, the growing area could range anywhere from 1 m^2 per crewmember (CM) for a simple "salad crop" setup, to more than 65 m^2/CM to provide a complete and varied diet. Mounting points must be provided in this area for lighting, air circulation, and irrigation hardware, and the distance from light to canopy (i.e. the "head height") is crop dependent. This head height requirement limits how much reconfiguration can happen after initial design since taller crops will not fit in shorter spaces. Finally, an access corridor (even if only temporary) must be provided for maintenance and, if there is no automation, for access so crops may be tended.

Lighting and growing surfaces may be arranged horizontally and, for some crops, vertically hanging arrangements can be used, especially when employing automation. Lighting fixtures must be arranged with a density such that their intensity is appropriate to the crop. Many crops like leafy greens grow best in lower intensity light while others such as tomatoes, wheat, maize, and rice can make use of very high intensity light [43]. Like with the head height, intensity limits possible later reconfiguration since dimming lights is much easier than increasing the number of fixtures, and low light areas will be less suited to growing higher light crops. Likewise, specialized lighting fixtures with the ability to shut off the red, blue, or far-red channels could be designed to allow growers more control over photosynthetic efficiency, trait selection, photomorphogenesis, or photoperiodic triggers.

Irrigation systems for BLSS tend to be strictly hydroponic, and the best researched variant is the nutrient film technique (NFT), which uses a thin layer of constantly running water to aerate and fertigate the root zone [44]. Another possible method is deep water culture (DWC), which uses a much thicker layer of water and must be aerated with a bubbling stone. Sleeve culture is a mid-way between DWC and NFT which lacks the large air to water interface of NFT, but requires much less circulating water than DWC. Hanging drip systems are a vertically oriented version of hydroponics, which combines a large air to water interface from constantly dripping water with a low circulating volume. The vertical orientation of the irrigation channels is also conducive to the integration of automation or movable shelving via overhead-hanging conveyor tracks. However, lighting and airflow must also be reoriented, and not all crops are suited to this orientation. Aeroponics has several advantages, such as a very low circulating volume and a potentially reduced root allocation from air pruning. However, this comes with the disadvantage of higher system capital and maintenance costs, due to the need to supply high pressure nutrient flow and to frequently replace the high pressure nozzles. All these systems have the additional requirement of needing power, pumps, and at least one storage reservoir for the recirculating nutrient solution. Special attention should be paid to making the system easily accessible for draining, cleaning, and maintenance.

A fertigation subsystem is also required for automating the addition of concentrated nutrient solution and acid, maintaining the recirculating solution's electrical conductivity (EC), and maintaining its potential of hydrogen (pH). This system usually involves two tanks of concentrated nutrient salts which are separated to prevent precipitation and are mixed, diluted, and injected into the circulating nutrient solution in order to maintain the appropriate concentration of nutrient ions, measured as EC, for uptake by the root zone. A third tank of concentrated acid is also needed for reducing the pH of the circulated nutrient solution. A subsystem of the fertigation system may possibly be connected to waste management so that sources of nitrogen (e.g. urea) or other nutrients may be imported into the growing system. In that situation, a biofilter subsystem may be used to provide surface area for nitrifying bacteria to colonize. Different crops have different optimal setpoints for EC and pH; therefore, either a compromise must be made between all grown crops or separate nutrient reservoirs and pumps must be used for the various crop roots zones.

Growing area HVAC is slightly more complicated than simply mixing crew air with that of the growing space because the humidity and temperature levels required for ideal plant growth are typically well above that which is comfortable for crew. Moreover, the psychrometrics of conditioning such air will not only require extra cooling to reach the dew point but a reheat step to ensure the outlet air is not too cold. For these reasons, the crew compartment HVAC should not be expected to supply the grow area HVAC, and it is likely a separate cooling (and reheating) system will be needed to supply air to the plant chamber. Moreover, this HVAC system will need to be connected to a network of distribution pipes throughout the growing area that will maintain adequate air speeds within the canopy. This circulation is critical to maintaining high levels of photosynthesis without causing pathological conditions. For instance, stagnant air in and around leaves will slow transpiration and gas exchange and can stall or disrupt growth, either by slowing photosynthesis directly or by preventing the flow of nutrients to the growing tips of leaves (as in tipburn) [45]. The growing spaces HVAC should also handle the mixing of crew air and growth zone air to ensure humidity and temperatures in both zones are maintained and that CO_2 (and perhaps sulfides, which plants may absorb through their leaves) is exchanged.

Likewise, the EMC of the growing area will have many plant-centric considerations and extra sensors which will require a separate control for the growing volume. Examples of what extra tasks a growth zone EMC will need to perform are: monitor light intensity, control lighting schedule, monitor pH, EC, & CO_2, monitor and control irrigation reservoir level, collect and analyze computer vision data, estimate biomass and gas exchange rates, and run modeling simulation to track against observed plant growth.

For proper closure, carbon (and other elements) must not only be absorbed into edible plant tissue but also released from inedible biomass [46]. Some kind of subsystem for efficient microbial breakdown, digestion, incineration, pyrolysis, combustion, or leaching of this biomass must be provided in the habitat, though it need not be in the plant growth space directly. During periods when CO_2 released from crew or reclaimed from biomass is not sufficient to maintain the required growth, additional CO_2 will need to be injected into the growing volume. Such a shortfall is quite likely to happen due to

unavoidable inefficiencies in the carbon cycle (such as air leakage or structural sequestration), especially when supplying all needed crew calories which necessarily requires more CO_2 than crew can respire. In contexts where there is local CO_2 (say from the Martian atmosphere), this CO_2 can be supplied by ISRU. Otherwise, it will need to be imported and stored cryogenically. Failure to supply make-up CO_2 will necessarily result in reduced growth rates, which will reduce total system energy efficiency and overall capacity.

Growing plants (and algae) produce many VOCs, some of which can directly affect or harm plant development (ethene and isoprene). Some VOCs may also build up over time potentially posing a hazard to crew or equipment. Removal of these trace organics can be generally achieved by high temperature catalytic combustion (such as via platinum coated glass-bead swing intake systems: like Swingtherm). A low temperature alternative is photocatalytic oxidation which uses much cheaper catalysis such as TiO2 (potentially with graphene substrate) and UV-C light to decompose compounds [47,48]. Additionally, compounds containing a double bond, such as ethene and isoprene can be removed by potassium permanganate coated granules (such as Purafil) [8]. The sparing requirements and replacement times for such catalysts in a BLSS are not yet well understood and would likely vary depending on what is grown.

In addition to VOC removal, it may occasionally become necessary in larger systems to remove O_2 during periods when there is an overproduction. This can occur when growing food for calories because there may be a delay (months) between the growing of tissue and in the ultimate freeing of carbon from inedible biomass: for instance, when an indeterminately grown plant is producing fruit. Equipment to remove excess oxygen is similar to that for CO_2 removal, consisting of zeolite beds specific to O_2 molecules which can concentrate the O_2 either for overboarding or compression and storage.

A sealed compartment must be set aside for nutrient salt, acid, substrate, and seed storage. Nutrient salts and acid will be needed for preparing replacement batches of hydroponic nutrient solution concentrate, while substrate and seeds will be needed for starting new growth. Due to their low mass, salts, substrate, and seed can likely be imported from Earth, at least for early BLSS. Later stage settlements should look for opportunities to produce salts, substrate, and acid locally through ISRU.

In later stage BLSSs, a special propagation system can be added for local production of seeds from flowers. For early BLSSs, the propagation area will simply be used to start new seedlings before transplant into the production system. Because seedlings use reduced strength nutrient solution, the propagation system will likely require a separate fertigation and irrigation system, but may piggy back on the existing growth zone HVAC.

A cleaning area/head-house (perhaps including a washing machine) must be provided for a BLSS. It is here that biofilms from microorganisms such as algae may be removed and equipment prepared for future production. The space can double as a head-house for the weighing, drying, and packaging of produce and the calibration of instruments.

More advanced BLSS configurations will also make use of computer vision [49] for monitoring, estimating, and providing feedback to growth models [50]. This system will require one or more cameras capable of imaging all or most of the growing area. Graphics software will then need to process these images to create a virtual model (or 'digital twin') of the growing plants. This feedback will help the EMC subsystem to autonomously evaluate crop status and size and better understand how much gas exchange is happening in the system.

Moving shelving is important for increasing the area to volume ratio and light use efficiency of the growing space. Moving shelves can also be connected to conveyor systems for automating movement of plants through the growing space. This can either improve worker efficiency by consolidating human activities to a work zone or by bringing plants to a robotic workstation. It can also improve the time-averaged growth uniformity by cycling the plants through any environmental anomalies present within the growing space.

A robotics system, if it is available, has the potential to allow tasks such as pollination, leaning and lowering, transplanting, harvesting, and cleaning to be done automatically. This type of time saving technology is actively being developed in vertical farms and is most developed in large commercial greenhouses.

3.3. ECLSS to BLSS integration requirements

Gas exchange, the removal of CO_2 and the creation of oxygen, is one of the most important roles of a BLSS. Provided it has sufficient growing area (around 20 m^2/CM), as plants grow to maturity in a BLSS, it can gradually take over as the primary ECLSS system. This means fewer spares will be needed for the PYC life support and less mass will need to be shipped as stored food for the crew. It is important to emphasize that although the BLSS becomes the primary system, many of the functions of the PC ECLSS, such as trace gas monitoring, are still critical to the system as a whole.

In order to prevent large fluctuations in system gas-exchange capacity, plants will need to be transplanted continuously, meaning plants of the same crop will be at varying levels of maturity. This scheduling can happen daily or can be staggered out into cohorts 5 to 7 per growth cycle. Software such as SIMOC [15,21] can be used to help plan this schedule and model how bioregenerative gas exchange will interact with the PYC system. This type of software will be critical not only for planning crop output and scheduling worker time but also for calculating how many spares the PYC ECLSS will need over the course of a mission.

Because most crops have significant inedible portions, a great deal more oxygen will be released than is needed when supplying most of the crew's calories from a BLSS. Harvest Index (HI) is the ratio of edible to total biomass produced when harvesting a mature crop. The HIs for NASA's highly researched exploration life support (ELS) crops range from 25 % to 90 %, averaging about 60 %. This means, less than 60 % of the CO_2 needed to grow food for the crew can be expected to be released from grown food through metabolic processes from the crew [46]. This reality naturally divides BLS into three categories: BLS for the sake of fresh food and flavors (e.g. BVAD's "Salad crops", 1 m^2/CM), BLS for O_2 production and CO_2 removal (crops with high HI are preferred so that more food is provided, ~20 m^2/CM), and BLS for total food production (which must have a variety of crops capable of providing a balance of vitamins and complementary proteins, >65 m^2/CM).

When used for O_2 renewal, photosynthesizing plants would offload the CO_2 removal task from zeolite/amine and hydroxide based filter systems. When the BLSS is in full operation, cabin CO_2 levels might be expected to be maintained around 800–1200 ppm in order to support optimal photosynthesis. After BLSS has taken over as the primary system, the PC systems may act in a diminished capacity to help balance CO2 and O2 concentrations in various compartments, both ensuring CO2 enrichment for plants and that O2 doesn't build up to dangerous levels [51].

Carbon dioxide is not the only gas which plants are capable of removing. Sulfur dioxide (oxidizes in water to hydrogen sulfate) and ammonia, urea, and nitrates can also be absorbed by leaves [52,53], and, thus, a BLSS can potentially act as a way to remove the output of catalytic combustion of sulfur and nitrogen compounds without having to deplete chemical filters. It is not clear how large a leaf area is sufficient to meet spacecraft maximum allowable concentrations (SMAC) or even if they can be fully achieved in this way.

Another function BLS may provide is water filtration. Plant root zones may directly absorb many types of ions, and, with the help of microorganisms, more complicated organic solutes can also be broken down and absorbed as nutrients. Although not necessary for the adequate growth of plants, the development of a culture of microbes is typically inevitable in the root zone of a hydroponic system. Sometimes these are merely a nuisance (such as some algaes, which may take some of the nutrients for themselves and create odors and biofilms),

occasionally they are harmful (such as pathogenic e. coli), but most often they are harmless or even helpful (as noted, by their ability to break down organics) [54]. In the farming industry, monitoring the microbial community for pathogens and nurturing a good culture are best practices.

When used in this way, the root zone of a BLSS can take over many of the functions of a VDC. Water taken up by the roots is purified as it passes through the plant and is, ultimately, emitted as transpiration from the leaves. The transpirate (which is largely, though not entirely water) can then be processed in a similar way as condensates from human sweat and breathing when pure water is needed. Much of it, however, can bypass that purification and be sent directly back into the irrigation reservoir, provided contamination from the cooling structure is not an issue.

The provision of fresh food has a psychological benefit which only a BLSS can provide. These crisp, fresh textures, aromas, and flavors cannot be replicated by stored food. Likewise, the aromatics produced by growing crops have been reported to help soothe and relax crew in the antechamber of the SPGC at Amundsen-Scott [55]. Though it is not clear if these benefits can be retained for the crew in the face of trace gas removal, it may be possible to maintain a "green space" or lounge area upstream of where filtration happens.

Despite the fact that stored food does not compare to fresh food on a hedonic level, it is still important to compare their masses in order to evaluate the potential for reducing the need for resupply from Earth and for extending shelf life of food stores.

According to NASA BVAD 2022, the nominal energy load for a crew member is 12,778 MJ/CM·day (3054 kcal/CM-·day), this includes 30 min of aerobic and 60 min of resistive exercise per day [1]. Given the reduced gravity of the moon and Mars, this exercise load is assumed to still be needed for most mission contexts in which a large BLSS is feasible (reduced gravity is present; stays are long enough to complete a crop cycle). Table 1 compares the effect of crop selection on growing area against NASA baseline BLSS diets and crew nutrition needs [1 pp. 172–176]. The growing area required for each crop selection is calculated from information from several tables, and calorie and nutrient information is extrapolated from USDA data [56], with the exception of cobalt, which is not present in hydroponically grown plants. Due to the need for redundancy in the food supply, it is highly likely that a store of shipped foods will be brought to a habitat even if locally grown foods are available. Because even the best preserved foods have a limited lifespan, this long-term supply will necessarily need to be integrated into the food supply and continuously turned over with 'fresh' resupply. Given previous validation of using 25 % outside-the-loop food from Bios-3 [4], a slightly more optimistic value of 20 % of crew calories was chosen for the supposed resupply turnover rate. Therefore, the baseline crop selection for this document was '20 % "as-shipped" food + extending "All ELS"', and the baseline growing area was 83.31 m^2/CM. Supplying outside food in this way can also help offset any nutrient deficiencies from a 100 % hydroponic diet. Particularly, NASA's "All ELS Crops" diet only provides 50 % of the daily recommended dosage of vitamin E and 66 % of calcium. Several other essential minerals are simply not present in (most) hydroponically grown plants. For example, cobalamin is fixed by soil based microorganisms before it is taken in by plants [57]. Moreover, hydroponic nutrient solution is often completely lacking in many essential and beneficial elements such as I, Co, Cr, Se, Si, Sr, and V [58,59]. Therefore, some form of supplementation (either in the hydroponic solution or in oral supplements) would be essential, even if 100 % of food was produced locally.

Table 2 compares the resupply mass of shipped vs locally grown foods. Because BLSSs require long stay times, it was assumed that the menu will need a greater variety than can be provided solely by dehydrated foods alone; therefore, NASA's "Menu Containing Some Frozen Food" [1] was used. This variety should help prevent dietary boredom which can make it more difficult for crews to hit their calorie targets. The nutrient recovery rate of 70 % is based partly on Lunn et al., 2017's [54] experiments with leaching, acid digestion, and microbial digestion of inedible biomass and also on closure rates reported by Bios-3 which experimented with recycled urine and reported 20 % mineral closure. It seems likely this number can be improved upon but more research is needed, especially on how much mass, power, and crew time would be needed for recovery systems. Long term integration tests for elemental buildup are also essential, as are tests on possible means of combating sodium buildup such as halophytes [60]. Ultimately, ISRU and soil extraction should be explored to relax the requirements for complete closure and provide a local supply of soluble nutrients.

The nearly 19:1 resupply mass reduction shown in Table 2 does not account for the extra mass of supporting equipment required by the growing system. To account for this, several tables were developed to explore the equivalent system mass (ESM) requirements for a growing system capable of providing 80 % of needed calories. Table 3 shows the mass of the growing system itself, its ESM power needs, its ESM thermal management needs, and its ESM volume containment needs. Together

Table 2
Food Resupply Mass For Shipped vs Locally Grown Food.

Shipped Food: Menu Containing Some Frozen Food		Grown Food Resupply Mass		Sources
	kg/ CM·d		kg/ CM·d	
Food "As-Shipped"	1.48	Initial Nutrient Salts	0.2546	BVAD, 2022 Tables 4–60 [1]
Packaging Mass	0.26	Recovery Rate[b]	70 %	
Locker Mass[a]	0.27	After Recovery	0.0764	Lunn et al., 2017 [54]
Freezer Mass[a]	0.866	Acid	0.0767	
Resupply Total	**2.876**	Resupply Total	**0.1530**	

[a] Based on ISS "pantry-style" storage.
[b] Conservative estimate, high uncertainty.

Table 1
Growing Area Needed vs Selection of Crops.

Growing Area Needed vs Selection of Crops					Sources
Crop Selection:	Total calories kcal/ CM·d	BLSS calories kcal/ CM·d	Area needed m^2/CM	Selection Provides:	Area ratio to 'All ELS' baseline	Derived from BVAD 2022, Table 4–89, 4–90, 4–91, 4–92 [1] and USDA nutrition labels [56].
NASA "ELS Salad Crops" diet	34.3	34.3	1.35	Limited fresh flavors	0.021	
NASA "Salad and Carbohydrate Crops" diet	1154.9	1154.9	19.50	Near 100 % gas exchange capacity	0.187	
NASA "All ELS Crops" diet	1914.7	1914.7	65.29	100 % nutrients except Vit. E, Ca, I, Co, Cr, Se, Si, Sr, and V	1.000	
Needed for metabolic target, extending "All ELS"	3054.0	3054.0	104.14	100 % local food	1.595	
20 % "as-shipped" food + extending "All ELS"	3054.0	2443.2	**83.31**	Redundant supply of packaged food	1.276	

Table 3
Equivalent system mass (non-crew time) for a large BLSS.

Grown Food Infrastructure Mass Capital Costs								Sources
Growing System Mass		*Power ESM*		*Thermal Management ESM*		*Chamber Volume ESM*		BVAD 2022, Tables 4–88 [1]
	kg/m^2		kW_e/CM		kW_{th}/CM		m^3/CM	
LED Light Fixtures[a]	4.10	Lights[a]	13.55	Heat-removed[f]	21.67	UnshieldVolume[g]	34.6	
LED Power Supply[a]	8.20	Heat-removal[c]	6.77					
Mechanization	4.1	Pumps & Fans[d]	2.03					BVAD 2022, Tables 3–16 [1]
Secondary Structure	5.7	Total	22.35					
Shoot Zone	3.6							
Root Zone[b]	2							
Water	from ISRU							BVAD 2022, Tables 3 and 4 [1]
Total	27.70							
Area Required	m^2/CM	PV Cell with Storage[e]	kg/kW_e	Mars Surface	kg/kW_{th}	Mars Surface	kg/m^3	
	83.31		178		146		9.16	
	kg/CM		kg/CM		kg/CM		kg/CM	
Mass	2307.99		3978.17		3164.11		316.94	
Total Mass	**9767.20**							

[a] Values based on "Heliospectra: Sierra" light bars targeting ~450 μmol/m^2·s at 2.8 μmol/J [62].
[b] Rough estimate from authors' experience in vertical farms.
[c] Assuming growing space HVAC dehumidifying chill/reheat cycle uses 50 % of lighting power.
[d] High estimate at 15 % of lights.
[e] Figure is roughly the same for Mars or Lunar context; storage type may vary.
[f] Assumes 5 % of total light energy absorbed as chemical energy by plants, all other electrical energy is converted to heat.
[g] Unshielded volume used, shielding assumed from ISRU shielding structure.

these are the non-crew time (NCT) ESM_{NCT} as defined in Levri et al., 2003 [61]. Crew Time (CT) ESM_{CT} is explored separately, in Table 4. For ESM_{NCT}, the current version of the BVAD lacks baseline information for LED lighting systems; therefore, modern, dimmable, single channel air-cooled grow lights with an efficacy of 2.8 μmol/J were used as an analog [62]. Even higher efficacy lights are already available, but they are not yet widely in use. Rather than account.

For variation of lighting intensity between crops in the growing area, a single, middling intensity of light was used: 450 μmol/m^2 as a general estimate. Root zone structural mass was not separated from water in the BVAD, therefore a rough estimate of 2 kg/m^2 was used. Water itself was assumed to be provided by ISRU. Due to the high humidity and latent heat load of a growing environment, the HVAC cycles for indoor cultivation have been shown to require substantial energy from reheat; therefore, the total energy for heat-removal is assumed to be roughly 50 % of the light energy [63].

A further 15 % of lighting power was assumed to be needed for air and nutrient solution circulation. A very small portion of energy spent in the farm is captured as chemical energy in the plants. In a CO_2 enriched growing environment, C3 photosynthesis becomes more efficient as photorespiration reduces; however, due to the assumption of multi-cropping, the exact efficiency is more complicated to calculate, therefore a middling number of 5 % was used [64]. The remaining energy must be removed from the growing space as heat by thermal management systems. Finally, the growing volume containment structure was assumed to be an unshielded container housed under a larger unpressurized shielding structure made from ISRU material (i.e. regolith or lava tube) where the rest of the habitat will also likely reside.

The mass figures calculated in Table 3 are strongly affected by power generation system ESM conversions (and the subsequent heat removal requirements). Also, these figures assume all power generated for the BLSS will be generated by dedicated systems. If lighter technologies are developed or if power and thermal management can be shared with or inherited from other large users (such as fuel ISRU systems), the ESM of the growing system could be substantially reduced.

Table 4 shows calculations for the ESM_{CT}. The table is broken down into automated and non-automated (manual) ESMs. The total time available for work was assumed to be 8 h per day. This was based on BVAD table values allocating "variably scheduled time" between 8.5 and 11.5 h/day on ISS. Of this, only 6.5 was actually allotted to operations. Given its common use as the average work day length in the US, 8 h seemed a reasonable mid-point, especially as maintaining extended work schedules during long duration missions would be difficult. Because automation practices are still under development, there is a lack of published data on exactly how much time can be saved and to what degree tasks are likely to be automated. Therefore, automation was assumed to reduce the time required to operate the BLSS by 66 %. This represents a 'best guess' by the authors based on labor studies in non-automated vertical farms which showed that harvesting, packing, and transplant time together can average over 70 % of labor time [64]. Using the methods described in Levri et al., 2003 [61], ESM_{total} was then calculated.

Table 5 breaks down the amount of time required to realize the benefits of reduced resupply mass offered by using a BLSS in the face of the increased ESM of installing the system in the first place. With the given set of assumptions, this is shown to be around 11.2 years with automation and 15.1 years without. Again, it should be noted that these numbers could be significantly less if power and thermal systems can be shared or if lighter power generation technologies are used, potentially removing several years from the return on investment (ROI) time. Also, longer work days (up to 11.5 h) can very slightly reduce the ESM_{CT}, saving about half a year in the automated case and 1.5 years in the non-automated case. Full ISRU production of nutrients and acids would save about 0.6 years. Increasing lighting efficacy, from 2.8 to 3.4 μmol/J,

Table 4
Equivalent system mass (crew time) for a large BLSS.

Crew Time ESM				Sources
Time for Operation of BLSS	0.03559		CM·h/m^2·d	BVAD 2022, Tables 4–88 [1]
Growing Area Required	83.3115		m^2/CM	
Total Work Time Available	8		CM·h/CM·d	
Automation Systems?	yes	no	66 % reduction[a]	BVAD 2022, Tables 3–28 [1]
Crew Time Spent on BLSS (t_{BLSS})	1.0082	2.9653	CM·h/CM·d	
Work Time Left Over ($t_{mission}$)	6.9918	5.0347	CM·h/CM·d	
$t_{BLSS}/t_{mission}$	0.1442	0.5890		
ESM_{NCT}	9767.2	9425.6	kg/CM	
ESM_{CT}	1408.4	5551.4	kg/CM	
ESM_{total}	**11,175.6**	**14,977.0**	kg/CM	

[a] Rough and conservative assumption, mass requirements uncertain.

Table 5
Time required for a large BLSS to reduce mission mass.

Mass Return on Investment (ROI) Estimates		
Food "As Shipped" (Re)supply Mass	2.876	kg/CM·d
Grown Food Resupply Mass	0.1530	kg/CM·d
1 year =	365.24	days
average E-M conjunction time	778.49	days
Total ESM for Automated Growing System	11,175.6	kg/CM
Time Required for ROI	4104.2	days
=	11.24	years
	5.27	Earth-Mars Conjunctions
Total ESM for Manual Growing System	14,977.0	kg/CM
Time Required for ROI	5500.3	days
=	15.06	years
	7.07	Earth-Mars Conjunctions

removes 1.4 years by reducing power and thermal ESM. If only the mass of the growing system was considered (no ESM considered at all), the ROI time could be as low as 2.3 years.

4. Future research required

Looking at both the background research and the system integration requirements provides a perspective on what research still needs to be done. Following are some suggestions on what direction future research should take.

4.1. Predictive modeling (long term and instantaneous)

Interfacing a BLSS with a PYC ECLSS is a challenge that will require new tools. As complex systems that have a great deal of variability within them, it will be important to be able to provide accurate information on the instantaneous intake and output of resources and the overall status of the BLSS (and all the types of plants within). Additionally, the system will need to be able to make predictions for long term planning. In order to provide accurate data to the ECLSS, validated growth models for each type of plant grown will be needed. To date, several mathematical models have been developed for some specific types of BLSS crops [1] in relevant environments. These and other mathematical models will need to be consolidated into a predictive modeling program (similar to SIMOC) that will provide both real time and long term predictions to the BLSS for adjusting the growing environment and for the crew to make decisions, such as when and what to plant. Moreover, several NASA crops have not been modeled, and any new crops will need fundamental research to create and validate predictive models for integration into BLSS subsystems. Integrating new approaches such as monochromatic day-length extension may also require updated models for older crops.

4.2. Plant specific data for models

Each plant has specific growth parameters and requirements. To be used in an integrated BLSS, extensive information is needed on the plants and their specific cultivars. One area of research is to look into how detailed this information actually needs to be for the models to provide accurate information to the BLSS. For instance, the modified energy cascade MEC model provided in the NASA BVAD lists 27 parameters used to calculate growth and gas exchange [1]. These parameters were based on a single baseline planting density and typically are from NFT experiments done in systems using high pressure sodium (HPS) lighting fixtures. Therefore, factors like planting density and spectral composition are not, and cannot be, reflected in these models. Should growth systems deviate from these baselines, which LED systems will necessarily do since they cannot produce the same spectrum as HPS lights and do not provide as much radiant heat at the canopy level, it is possible that model accuracy will be reduced. In order to get accurate modeling parameters, these plant specific and environment specific factors need to be ground validated in large, closed growing systems with integrated crops (all crops grown together) using the relevant technologies.

4.3. Sensors and feedback

Like the plants themselves, models cannot function in a vacuum, a model must have feedback on the current plant status. Otherwise, growth may deviate from expectations, and the system would have no way of adjusting to the changes. The CEA research community is developing sensors which can be used for making biomass predictions or detecting stress states in crops. For instance, computer vision can be used to measure factors such as plant height, top projected canopy area (TPCA), and spectral information [50,65,66] to inform growers (and potentially computer control systems) how the plants are growing. This type of feedback system will be an important part of integrating automation into the growing environment.

4.4. Labor tracking and saving

Astronaut time is one of the most valuable resources that a crew on another celestial body will have, and transplanting, harvesting, and caring for crops are particularly time consuming tasks. In addition to labor spent on the plants, tracking the labor required to maintain, operate, calibrate, and repair a BLSS is a vital area of research. While this has been looked at in the past, as mentioned in section 3.3, there remains much work to be done in identifying time consuming activities. This information will also inform efforts to develop automation technologies to save crew time. Moreover, these technologies will apply directly to the vertical farming industry on Earth and can be used to save on labor costs, which is the greatest cost in that industry [67,68].

4.5. Robotics and automation

Automation of common tasks is actively being developed in the CEA industry; however, there are additional constraints when considering automation for BLSS. The hardware must be robust and accessible for cleaning and maintenance while at the same time being lightweight and compact. These engineering requirements are often at odds with each other. As shown in Table 4 the potential mass savings of automation are large, representing a 34 % reduction in ESM and a 3.8 year savings on ROI. Therefore, research into systems for automation of the BLSS systems is a priority.

Examples of types of automation which need more research and development include: moving gutter/shelving systems; automated harvesting; robotic arms for pollination, leaning & lowering, pruning and transplant activities; and computer vision technologies for generating 3D representations of plants and measuring factors such as canopy depth, plant architecture, spectral outputs, and plant status. Because the vertical farming industry currently focuses almost exclusively on leafy greens, any technologies for automating other types of crops will have application on Earth for extending the types of crops which may be grown in vertical farms and possibly also greenhouses.

4.6. Volatiles, cycling, and filtration

Research into entirely closed agricultural systems has been incredibly limited. Even tightly closed vertical farms tend to have leakage rates high enough to prevent trace gasses from affecting growth. Completely closed atmosphere systems have additional risks from such gas build-ups, and research on the impact of this build up on plants as well as the methods needed to counteract it remains limited. Likewise, few farms choose to continually recycle nutrients for long periods. As such, the effects of recycling of elements on nutrient composition has not been researched sufficiently. For this research to be done, large (walk-in scale), high fidelity enclosures must be built for growing plants in long,

continuous production cycles. Ideally, multi-cropping should be done to check for cross-species interaction, as happened in Bios-3 when researchers connected the atmospheres of algae chambers to plant chambers and discovered a negative impact on plant growth [2]. Sodium free (probably K based) hygiene products (e.g. toothpaste and soap) alternatives should be developed and tested for their ability to help prevent Na build-up.

The ability of plant leaf surfaces to absorb trace contaminants needs to be better understood. Research should be done to explore how much leaf surface is needed to reach SMAC for various trace gasses. Likewise, the root zone of plants is an exceptional regenerative filter for removing water-based contaminants. More study is needed to integrate nutrient recycling techniques with solid and brine waste management. The ESM for such systems needs to be baselined, and the total regenerative capacity of the root zone using various techniques (biodigesters, leaching, incineration, etc.) needs to be characterized.

There also exists possible synergies between PYC contaminate removal and growth chamber HVAC systems. Because of the reheat requirement of growth chamber HVAC, it is possible to do hot catalyst trace gas combustion inline with the dehumidified return air stream. Doing so allows any waste heat to contribute to the reheat process and outputs combustion products straight to the growing space for filtration. This possibility should be explored to see if it has meaningful levels of removal capacity.

4.7. Sparing and ISRU

Operation of a BLSS will reduce the amount of sparing needed for a PYC ECLSS, but will also create new sparing requirements of its own. The most massive element needing replacement in air filtration systems are the activated charcoal beds, which can become saturated over time. The particle and bacteria filters may also become saturated, but they are somewhat regenerable by periodically vacuuming [69]. Eventually, catalysts may also need replacement. Both sulfur species that produce sulfur dioxide or hydrogen sulfide and nitrogen species such as ammonia and ammonium, can act as irreversible poisons to catalytic surfaces, forming strong Pt-XO or Pt–O bonds on the surface [11,12]. The time required for replacement is highly dependent on the expected contaminant load, so it is critical to understand what volatile loads from crops, root zone related microorganisms, food preparation, cooking, and rot will be produced in an integrated BLSS environment. All of this testing requires high fidelity enclosures. Likewise, as stated in section 4.6, the ability of the canopy to filter such contaminants needs to be characterized so that the impacts on filter sparing can be calculated.

Water filtration sparing is equally impacted. Physicochemical filtration often involves not only activated charcoal, but also a series of heavy resin beds which can become depleted. None of these filtration technologies is regenerative, and sparing can represent a substantial mass [30,70,71]. Transpiration has the potential to greatly reduce sparing requirements by providing a means of bioregeneration. The interplay between water filtration and the root zone needs more research so that new sparing lifetimes for the PYC side can be calculated.

One way to further reduce the shipped mass required by BLSS is to use ISRU to create salts and/or hydroponic nutrient solution locally. Research should be done to explore methods for extracting nutrient salts from ISRU sources. Particular attention should be paid to extraction of nitrogen salts as this is the majority of the needed shipped mass (~60 %). In total the salts of nitrogen and the other macronutrients (K, Ca, Mg, P, and S) would constitute around ~99 % of the mass of the needed salts, with salts of the micronutrients (Fe, B, Mn, Zn, Cu, Mo) making up the rest. Local acid production, while possibly hazardous and energy intensive (depending on the scale and the process used), might also be explored as a means of completely removing the need for nutrient resupply. Nutrient solution formulations with more ammonium may also help reduce the nitric acid requirements and should also be researched [72], with special regard to multicropping.

Typically, plants are grown on some kind of substrate (a material used for physically supporting the roots as they grow). Some examples include rockwool, coco coir, and fleece. This is an element of the production process which is not properly baselined in the BVAD. While potentially quite light, the mass of used substrate can add up over time and produces a waste stream which may not always be recyclable. Therefore, substrate recycling and ISRU technologies should be explored, and different types of substrate (and the equipment needed to prepare them) should be characterized and baselined.

4.8. Propagation

The final reduction of resupply mass requires the creation of a local source of seeds. This implies a growing space set aside for flower production. Because photoperiodic flowering triggers may require different durations of lighting and temperature, this growing space may need its own zone or its own separate growing system. The crew time required to operate the system needs to be baselined for each crop type, and the area requirements to produce sufficient seed needs to be characterized relative to the food production area.

Although plants have been shown to be extremely tolerant to gamma radiation, the effect of heavy ions in cosmic radiation on seed, seed storage, seed formation, and germination remain unclear. Will propagation without radiation shielding on the surface maintain genetic fidelity? Can plants be grown in vacuum surrounded greenhouses without artificial lighting? These questions should be answered.

There are also other means of propagation for many species of plants. Strawberries produce runners, and stems of many plants will develop adventitious roots. Another means of clonal reproduction is callus culture. The ESM requirements of these methods should be baselined, and potential crop species which are amenable to them should be identified and listed.

There has been limited research on potential space pollinators. University of Guelph, for instance, studied bumble bees (*Bombus impatiens*). However, difficulties in the logistics of transporting honey or bumble bees through a long transit, as well as other cultural concerns, could make using them more challenging. Other species of bee may be more suited, such as the mason bee (*Osmia lignaria*), a solo bee that hibernates and can be more easily stored long term. Breeding programs to produce lineages with characteristics favorable to space flight should be studied and created. Characteristics to look for are: non-aggressive, capable of long hibernation (for transit and storage), short range habits (does not fly far for pollen), works well with circulation systems (won't attempt to nest in inappropriate places).

Robotic pollination is an alternative which may also be explored. To support such technology, computer vision systems will also need to be developed for identifying flowers and other plant parts. Many flowering practices rely heavily on vegetative/reproductive steering and fruit load management, so vision systems should also be able to measure and identify important anatomical features such as internode distances, number of leaves, number of fruit, and meristem locations. This technology is fairly species specific, and, for some crops, it is already somewhat developed, and further development would have many applications on Earth.

4.9. Food preservation and processing

Harvesting is typically as simple as separating edible from inedible biomass when plants have matured. More complicated growing techniques allow multiple harvests from the same plant and take slightly more care. All harvesting invariably creates a resource stream of fresh food which must be processed in some way to extend its shelf-life, preserve flavor and texture, and retain moisture. Much hand waving exists in the literature stating that the output of the system will go straight to the table and so packaging requirements are minimal. However, practical considerations dictate that some degree of storage

may be required to accommodate a rotating menu, where not all grown foods are used every day. Moreover, the creation of a supply of locally preserved food could help to further off-set the need for resupply and hedge against the risk of shipping losses. Additionally, some foods are not always suitable for use immediately after harvest. Grains, for instance, must be ground into a powder for many cooking applications. Bran must be separated from the germ. Oils must be pressed from seed. Many ferments and sauces require a vessel. The ESM needs of the equipment for preserving and processing growing system output has not been properly baselined. Design options should be studied and the needed crew time measured for the various methods of preparation.

Disposal of inedible biomass is also an inevitable part of growing crops. Although recycling is the final fate of this material, there are some preprocessing steps which should likely occur before it is handed off to a recycling system. First is measurement. The ESM of equipment needed to perform scientific and engineering analysis on harvested biomass such as scales, cameras, and other sampling equipment needs to be baselined. Next is drying (and then even more measurement), the ESM of drying and or dehydrating equipment also needs to be baselined. This will help with taking accurate biomass measurements for study, with reclaiming water, and with reducing the mass of the output stream to the recycling system.

On many of the worlds in our solar system, there are opportunities for using preservation techniques which are not easily done on Earth. Vacuum for instance is much easier to make on the moon and Mars. Availability of argon gas may be much higher on Mars, where CO_2 concentration processes or nitrogen harvesting will likely also capture a large amount of this inert gas. Reduced pressure environments may also be realized as habitat designers try to minimize structural loads [33–35]. These alternate atmospheres will potentially allow easy access to unique preserving environments for long term food preservation. New techniques for food preservation using alternate atmospheres should be studied.

Another area requiring study is to characterize off-gassing (including humidity) from stored foods (ethene particularly), from cooking processes, from fermenting, and from rot. What happens in contingency scenarios where refrigeration is lost or a section of crop dies? What are the filtration requirements for threshing, grinding, and milling processes? What are the ESM requirements of canning or other long term food storage?

4.10. Additional BLSS modules and their integration

Research is being conducted on BLSS modules that grow things other than plants. For instance, mushrooms, algae, bioreactors, insects and other products are areas of ongoing research that can expand the palette of available outputs. Moreover, the HI of many existing ELS crops includes only the traditionally harvested portions. Several of these crops have additional edible portions such as their leaves. These edible portions need to be accounted for in their baselines.

Additionally, new crop types should be added to the list of crops already baselined. Fundamentally, this will involve creating new growth models, establishing best cultural practices, finding optimal lighting intensities and photoperiods, assessing potential for monochromatic light extension, finding optimal CO_2, and baselining yield rates, HI, and gas exchange rates. More advanced research would then be needed in automation practices and computer vision methods for those crops. As new crops are integrated, new menus need to be made using the resource exchange and modeling software.

All systems within a BLSS need to be integrated and ground tested in tightly closed systems with the latest technologies and baseline diets. This will help characterize operational inputs and test for cross species interactions. Other spaceflight technologies that have the potential to emit gasses or affect water quality also should be tested in such chambers (or at least their emission profiles measured and duplicated) to verify there are no unexpected bio/technological interactions.

4.11. Lighting systems

In the near term, development should be done to create a set of lights specifically for BLSS that have high efficacy and that may run directly from a solar and battery storage DC power bus. This will remove both the mass and the electrical inefficiencies caused by inverters. These lights should be multichannel to allow various parts of the spectrum to be toggled as the grower wishes. Multi-dimming of channels is likely to be more costly and more massive, so it should not be a goal of this development. Rather, dimming should either dim all channels or just the red channel, which should represent the majority of photons. This second option, along with channel toggling, will also still allow the grower to adjust ratios such as red to far-red or red to blue, which are important for some physiological triggers. Baseline mass and power usage values should be generated for this lighting.

4.12. Artemis

NASA's Artemis program, which proposes to test long-term exploration systems, needs to begin testing of smaller scale BLSS systems so that the technology readiness level of BLSS can be advanced. Presently, mission surface stay times for Artemis are too short (30 days) to complete full crop cycles, even on the faster growing crops. Automation technologies to perform some of the early seeding and transplanting remotely could act as a work-around for this. Human presence would likely be an advantage during harvest times. In preparation for this, computational fluid dynamic analysis of irrigation flow should be performed to better understand the impact of lunar gravity on the NFT technologies most BLSS research has been based upon.

5. Conclusions

The only way to support human exploration of space in the long term is to develop BLSS technologies. Shelf-life and launch mass issues limit the utility of shipped foods to missions less than a decade. If we are to establish permanent habitats on other worlds, there is much still to be done in order to achieve the integrability, reliability, and functionality that will be required.

Fully closed growing systems need to be constructed so that crops can be baselined in the presence of relevant technologies. The trace gasses produced by those crops need to be understood so that the sparing requirements of combined BLSS and ECLSS systems can be calculated. The effect of trace contaminants on food production and the interaction of crops with contaminant removal and nutrient cycling needs to be better understood. Finally, automation technologies need to be developed for reducing the amount of crew time needed to operate BLSS systems. These are the highest priority research areas.

Most of these technologies will also have immediate implications for the vertical farm and greenhouse industries on Earth. Licensing of these technologies could prove a useful means of producing a return on the required investments. Moreover, the fundamental research on plant cultivars will form a body of data and mathematical models which could potentially also be licensed. This approach has already been proposed by some BLSS research and development firms.

CRediT authorship contribution statement

Megan Kane: Conceptualization, Project administration, Supervision, Writing – original draft, Writing – review & editing. **K.C. Shasteen:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] M.K. Ewert, T.T. Chen, C.D. Powell, Life Support Baseline Values and Assumptions Document, National Aeronautics and Space Administration, 2022.

[2] R.M. Wheeler, Agriculture for space: people and places paving the way, Open Agriculture 2 (1) (2017), https://doi.org/10.1515/opag-2017-0002.

[3] E. Nardone, P.G. Kevan, M. Stasiak, M. Dixon, Atmospheric pressure requirements of bumblebees (bombus impatiens) as pollinators of lunar or martian greenhouse grown food. Gravitational and Space Biology Bulletin, publication of the American Society for Gravitational and Space Biology, 2012.

[4] F.B. Salisbury, J.I. Gitelson, G.M. Lisovsky, Bios-3: siberian experiments in bioregenerative life support, Bioscience 47 (9) (1997) 575–585, https://doi.org/10.2307/1313164.

[5] Y. Fu, L. Li, B. Xie, C. Dong, M. Wang, B. Jia, L. Shao, Y. Dong, S. Deng, H. Liu, G. Liu, B. Liu, D. Hu, H. Liu, How to establish a bioregenerative life support system for long-term crewed missions to the moon or Mars, Astrobiology 16 (12) (2016) 925–936, https://doi.org/10.1089/ast.2016.1477.

[6] R.M. Wheeler, Gas-exchange measurements using a large, closed plant growth chamber, Hortscience 27 (7) (1992) 777–780, https://doi.org/10.21273/HORTSCI.27.7.777.

[7] C. Wehkamp, M. Stasiak, J. Lawson, N. Yorio, G. Stutte, J. Richards, R. Wheeler, M. Dixon, Radish, L. Cv. sativa, Cherry bomb II) growth, net carbon exchange rate, and transpiration at decreased atmospheric pressure and/or oxygen, Gravitational and Space Research 26 (1) (2012) 3–16.

[8] J.H. Batten, B.V. Peterson, E. Berdis, E.M. Wheeler, Biomass Production Chamber Air Analysis of Wheat Study (BWT931), 1993.

[9] R.M. Wheeler, B.V. Peterson, G.W. Stutte, Ethylene production throughout growth and development of plants, Hortscience 39 (7) (2004) 1541–1545, https://doi.org/10.21273/HORTSCI.39.7.1541.

[10] J.H. Batten, G.W. Stutte, R.M. Wheeler, Volatile organic compounds detected in the atmosphere of NASA's biomass production chamber, Adv. Space Res. 18 (4–5) (1996) 189–192, https://doi.org/10.1016/0273-1177(95)00876-G.

[11] Kh M. Minachev, N.I. Shuikin, I.D. Rozhdestivenskaya, Poisoning of platinum catalysts with a low content of active metal on a carrier, under conditions of dehydrogenation catalysis, Russ. Chem. Bull. 1 (4) (1952) 567–575, https://doi.org/10.1007/BF01164919.

[12] C. Özyalcin, P. Mauermann, S. Dirkes, P. Thiele, S. Sterlepper, S. Pischinger, Investigation of filtration phenomena of air pollutants on cathode air filters for PEM fuel cells, Catalysts 11 (11) (2021) 1339, https://doi.org/10.3390/catal11111339.

[13] V. Maiwald, V. Vrakking, P. Zabel, D. Schubert, R. Waclavicek, M. Dorn, L. Fiore, B. Imhof, T. Rousek, V. Rossetti, C. Zeidler, From ice to space: a greenhouse design for moon or Mars based on a prototype deployed in Antarctica, CEAS Space Journal 13 (1) (2021) 17–37, https://doi.org/10.1007/s12567-020-00318-4.

[14] SAM – SAM. https://samb2.space/sam/. Accessed September. 14, 2023.

[15] National Geographic. SIMOC. https://ngs.simoc.space/. Accessed November. 11, 2021..

[16] C. Verseux, M. Baqué, K. Lehto, J.-P.P. de Vera, L.J. Rothschild, D. Billi, Sustainable life support on Mars – the potential roles of cyanobacteria, Int. J. Astrobiol. 15 (1) (2016) 65–92, https://doi.org/10.1017/S147355041500021X.

[17] T.P. Ramalho, G. Chopin, L. Salman, V. Baumgartner, C. Heinicke, C. Verseux, On the growth dynamics of the cyanobacterium anabaena sp. PCC 7938 in martian regolith, NPJ microgravity 8 (1) (2022) 43, https://doi.org/10.1038/s41526-022-00240-5.

[18] M. Nelson, M. Finn, C. Wilson, B. Zabel, M. van Thillo, P. Hawes, R. Fernandez, Bioregenerative recycling of wastewater in Biosphere 2 using a constructed wetland: 2-year results, Ecol. Eng. 13 (1–4) (1999) 189–197, https://doi.org/10.1016/S0925-8574(98)00099-8.

[19] C. Lasseur, Melissa: the European Project of a Closed Life Support System, 2008.

[20] S.C. Gellenbeck, A Predictive Model for the Production Rates of a Bioregenerative Life Support System, Master thesis, 2018.

[21] Gellenbeck, S. Model and Design of a Fully-Integrated Bioregenerative Life Support System (FI-BLSS) Using Scalable Interactive Model of an Off-World Community (SIMOC). ProQuest. https://www.proquest.com/openview/5b89cd887df6e7682b1affaf7f0619e6/1.pdf?pq-origsite=gscholar&cbl=18750&diss=y. Accessed September. 11, 2023..

[22] J.H. Chung, Carbon Dioxide Generation by Fungal Mycelium during Spawn Run, University of Arizona, 2021. Master thesis.

[23] C.A. Mitchell, M.P. Dzakovich, C. Gomez, R. Lopez, J.F. Burr, R. Hernández, C. Kubota, C.J. Currey, Q. Meng, E.S. Runkle, C.M. Bourget, R.C. Morrow, A. J. Both, Light-emitting diodes in horticulture, in: J. Janick (Ed.), Horticultural Reviews, vol. 43, Wiley, 2015, pp. 1–88.

[24] M. SharathKumar, E. Heuvelink, L.F.M. Marcelis, W. van Ieperen, Floral induction in the short-day plant Chrysanthemum under blue and red extended long-days, Front. Plant Sci. 11 (2020) 610041, https://doi.org/10.3389/fpls.2020.610041.

[25] Y. Zhang, M. Kacira, L. An, A CFD study on improving air flow uniformity in indoor plant factory system, Biosyst. Eng. 147 (2016) 193–205, https://doi.org/10.1016/j.biosystemseng.2016.04.012.

[26] E. Peiro, A. Pannico, S.G. Colleoni, L. Bucchieri, Y. Rouphael, S. De Pascale, R. Paradiso, F. Gòdia, Air distribution in a fully-closed higher plant growth chamber impacts crop performance of hydroponically-grown lettuce, Front. Plant Sci. 11 (2020) 537, https://doi.org/10.3389/fpls.2020.00537.

[27] C.J. Kaufmann, Reducing Tipburn in Lettuce Grown in AnIndoor Vertical Farm: Comparing the Impact OfVertically Distributed Airflow vs. HorizontallyDistributed Airflow in the Growth of Lactuca Sativa. Master Thesis, University of Arizona, 2023.

[28] Heating, Ventilating, and Air Conditioning (HVAC) for Indoor Plant Environments without Sunlight.".

[29] S. Hemming, F. de Zwart, A. Elings, A. Petropoulou, I. Righini, Cherry tomato production in intelligent greenhouses-sensors and AI for control of climate, irrigation, crop yield, and quality, Sensors 20 (22) (2020), https://doi.org/10.3390/s20226430.

[30] M.J. Kayatin, D.L. Carter, R.G. Schunk, J.M. Pruitt, Upgrades to the ISS water recovery system. Presented at the 46th International Conference on Environmental Systems, 2016.

[31] G.L. Douglas, S.R. Zwart, S.M. Smith, Space food for thought: challenges and considerations for food and nutrition on exploration missions, J. Nutr. 150 (9) (2020) 2242–2244, https://doi.org/10.1093/jn/nxaa188.

[32] K.C. Takada, A.E. Ghariani, Van Keuren, Advancing the oxygen generation assembly design to increase reliability and reduce costs for a future long duration mission, NASA Technical Reports Server (2015).

[33] J.E. Herlocher, D.G. Quigley, V.S. Behar, E.G. Shaw, B.E. Welch, Physiologic response to increased oxygen partial pressure. i. Clinical observations, Aero. Med. 35 (1964) 613–618.

[34] W.G. Robertson, J.J. Hargreaves, J.E. Herlocher, B.E. Welch, Physiologic response to increased oxygen partial pressure. Ii. Respiratory studies, Aero. Med. 35 (1964) 618–622.

[35] R. Zalusky, F. Ulvedal, J.E. Herlocher, B.E. Welch, Physiologic response to increased oxygen partial pressure. Iii. Hematopoiesis, Aero. Med. 35 (1964) 622–626.

[36] Perry, J. L. Trace Contaminant Control for the International Space Station's Node 1 : Analysis, Design, and Verification. NTRS - NASA Technical Reports Server. https://ntrs.nasa.gov/citations/20170005170. Accessed September. 16, 2023..

[37] Jaunsen, J. R. The Behavior and Capabilities of Lithium Hydroxide Carbon Dioxide Scrubbers in a Deep Sea Environment. Defense Technical Information Center. https://apps.dtic.mil/sti/citations/ADA216268. Accessed September. 17, 2023..

[38] Tatara, J. D., Perry, J. L., and Franks, G. D. Overview of the International Space Station System Level Trace Contaminant Injection Test. NASA Technical Reports Server. https://ntrs.nasa.gov/citations/19980058812. Accessed September. 14, 2023..

[39] Cmarik, G. E., Peters, W. T., and Knox, J. C. 4-Bed CO2 Scrubber – From Design to Build. Texas Tech University Libraries. https://ttu-ir.tdl.org/bitstream/handle/2346/86440/ICES-2020-178.pdf. Accessed September. 16, 2023..

[40] Ranz, H., Dionne, S., and Garr, J. A Thermally-Regenerated Solid Amine CO2 Removal System Incorporating Water Vapor Recovery and Ullage Air Recovery. Texas Tech University Libraries. https://ttu-ir.tdl.org/bitstream/handle/2346/84942/ICES-2019-174.pdf. Accessed September. 17, 2023..

[41] J. James Lee Broyan, L. Shaw, M. Mckinley, C. Meyer, M.K. Ewert, W.F. Schneider, M. Meyer, G.A. Ruff, A.C. Owens, R.L. Gatens, NASA Environmental Control and Life Support Technology Development for Exploration: 2020 to 2021 Overview, 2021.

[42] J. Silverman, A. Irby, T. Agerton, Development of the crew dragon ECLSS. Presented at the International Conference on Environmental Systems, 2020.

[43] B.G. Bugbee, F.B. Salisbury, Exploring the limits of crop productivity, Plant Physiol. 88 (3) (1988) 869–878, https://doi.org/10.1104/pp.88.3.869.

[44] A. Cooper, Abc of Nft: Nutrient Film Technique: the World's First Method of Crop Production without A Solid Rooting Medium (181p), Intl Specialized Book Services, London, 1979.

[45] E. Heuvelink, Wageningen University, The Netherlands. Achieving Sustainable Greenhouse Cultivation, Burleigh Dodds Science Publishing, 2019.

[46] R.M. Wheeler, Carbon balance in bioregenerative life support systems: some effects of system closure, waste management, and crop harvest Index, Adv. Space Res. 31 (1) (2003) 169–175, https://doi.org/10.1016/S0273-1177(02)00742-1.

[47] J. Perry, K. Frederick, J. Scott, D.A. Reinemann, Comparison of photocatalytic oxidation reactor performance for spacecraft cabin trace contaminant control applications, in: Presented at the 41st International Conference on Environmental Systems, 2011. Reston, Virigina.

[48] K. Owen, Evaluation of a graphene enhanced photocatalytic oxidation device, International High Performance Buildings Conference (2022).

[49] A.S. Petropoulou, B. van Marrewijk, F. de Zwart, A. Elings, M. Bijlaard, T. van Daalen, G. Jansen, S. Hemming, Lettuce production in intelligent greenhouses-3D imaging and computer vision for plant spacing decisions, Sensors 23 (6) (2023), https://doi.org/10.3390/s23062929.

[50] K.C. Shasteen, M. Kacira, Predictive modeling and computer vision-based decision support to optimize resource use in vertical farms, Sustainability 15 (10) (2023) 7812, https://doi.org/10.3390/su15107812.

[51] J.H. Eley Jr., J. Myers, Study of a photosynthetic gas exchanger, a quantitative repetition of the priestley experiment, Tex. J. Sci. Legacy CDMS (1964) 296–333.

[52] T. Elkiey, D.P. Ormrod, B. Marie, Foliar sorption of sulfur dioxide, nitrogen dioxide, and ozone by ornamental woody Plants1, Hortscience 17 (3) (1982) 358–360, https://doi.org/10.21273/HORTSCI.17.3.358.

[53] M. Agrawal, Plant responses to atmospheric sulphur, in: Y.P. Abrol, A. Ahmad (Eds.), Sulphur in Plants, Springer Netherlands, Dordrecht, 2003, pp. 279–293.

[54] G. Lunn, G. Stutte, L. Spencer, M. Hummerick, L. Wong, R. Wheeler, Recovery of Nutrients from Inedible Biomass of Tomato and Pepper to Recycle Fertilizer, 2017.

[55] R.L. Patterson, G.A. Giacomelli, P.D. Sadler, Resource and production model for the South Pole food growth chamber, in: International Conference on Environmental Systems, No. 1, 400 Commonwealth Drive, 2008. Warrendale, PA, United States.

[56] National Agricultural Library, FoodData central. Ag Data Commons, 2019, https://doi.org/10.15482/usda.adc/1504533.

[57] H. Fang, J. Kang, D. Zhang, Microbial production of vitamin B12: a review and future perspectives, Microb. Cell Factories 16 (1) (2017) 15, https://doi.org/10.1186/s12934-017-0631-y.
[58] R. Nieder, D.K. Benbi, F.X. Reichl, Microelements and their role in human health, in: Soil Components and Human Health, Springer Netherlands, Dordrecht, 2018, pp. 317–374.
[59] D. Rehder, Vanadium. Its role for humans, Metal ions in life sciences 13 (2013) 139–169, https://doi.org/10.1007/978-94-007-7500-8_5.
[60] N.A. Tikhomirova, S.A. Ushakova, N.P. Kovaleva, I.V. Gribovskaya, A. A. Tikhomirov, Influence of high concentrations of mineral salts on production process and NaCl accumulation by salicornia europaea plants as a constituent of the LSS phototroph link, Adv. Space Res. 35 (9) (2005) 1589–1593, https://doi.org/10.1016/j.asr.2005.01.055.
[61] J.A. Levri, J.W. Fisher, H.W. Jones, A.E. Drysdale, M.K. Ewert, A.J. Hanford, J. A. Hogan, J.A. Joshi, DavidA. Vaccari, Advanced Life Support Equivalent System Mass Guidelines Document, NASA, 2003. Publication NASA/TM-2003-212278.
[62] Heliospectra. Heliospectra SIERA Technical Specifications V2.1. assets-global. website-files.com. https://assets-global.website-files.com/5ef8b3d80f5e291e40354f19/5efc54cd617d8d964b2827ae_Heliospectra%20SIERA%20Technical%20Specifications%20V2.1.pdf. Accessed September. 17, 2023..
[63] Sabeh, N., Ross, D., Miner, L., and Everett, S. Literature Review of Energy and Water Use in Controlled Environment Horticulture and Potential Efficiency Opportunities. Emerging Technologies Coordinating Council. https://www.etcc-ca.com/reports/literature-review-energy-and-water-use-controlled-environment-horticulture-and-potential. Accessed September. 17, 2023..
[64] K. Ohyama, J. Yamaguchi, A. Enjoji, Evaluating labor productivity in a plant production system with sole-source lighting: a case study, HortTechnology 28 (2) (2018) 121–128, https://doi.org/10.21273/HORTTECH03886-17.
[65] D. Story, M. Kacira, C. Kubota, A. Akoglu, L. An, Lettuce calcium deficiency detection with machine vision computed plant features in controlled environments, Comput. Electron. Agric. 74 (2) (2010) 238–243, https://doi.org/10.1016/j.compag.2010.08.010.
[66] D. Story, M. Kacira, Design and implementation of a computer vision-guided greenhouse crop diagnostics system, Mach. Vis. Appl. 26 (4) (2015) 495–506, https://doi.org/10.1007/s00138-015-0670-5.
[67] Agrylist, State of indoor farming, Arte (2017). https://artemisag.com/state-of-indoor-farming-new-2017/. (Accessed 17 September 2023).
[68] Agritecture. Global Controlled Environment Agriculture Census, Agritecture (2019). https://www.agritecture.com/census. (Accessed 17 September 2023).
[69] Perry, J. L. International Space Station Bacteria Filter Element Service Life Evaluation. NASA Technical Reports Server. https://ntrs.nasa.gov/citations/20050215100. Accessed September. 17, 2023..
[70] Orourke, M. J. E., Perry, J. L., and Carter, D. L. A Water Recovery System Evolved for Exploration. NASA Technical Reports Server. https://ntrs.nasa.gov/citations/20060047781. Accessed September. 17, 2023..
[71] Carter, D. L., Tabb, D., and Perry, J. Performance Assessment of the Exploration Water Recovery System. NASA Technical Reports Server. https://ntrs.nasa.gov/citations/20090001281. Accessed September. 17, 2023..
[72] R.M. Wheeler, J.C. Sager, W.L. Berry, C.L. Mackowiak, G.W. Stutte, N.C. Yorio, L. M. Ruffe, Nutrient, acid and water budgets of hydroponically grown crops, Acta Hortic. 481 (1999) 655–662, https://doi.org/10.17660/ActaHortic.1999.481.78.